

JSS 1-3

TEACHER PRACTICAL ACTIVITY MANUAL

Implementation guide for PA-301 to PA-400

QUESTION -> TEST -> MEASURE -> ANALYZE -> MANAGE

Purpose	Turn the GrowMeal practical cards into safe, repeatable, age-appropriate teaching sessions.
Teacher role	Prepare the site, demonstrate technique, supervise safety, question learners, check evidence and connect the activity to the journal, mission cards and assessment system.
Core cycle	PREPARE -> DEMONSTRATE -> PRACTISE -> RECORD -> DISCUSS -> CONNECT -> CLEAN UP
Assessment	Use observation, student records and the linked practical/investigation evidence. Avoid grading recall alone.
Local adaptation	Substitute locally available materials when the learning objective, safety and measurement quality are preserved.

WHOLE-MANUAL SAFETY & IMPLEMENTATION STANDARD

Teachers must inspect the garden before each activity; establish boundaries; match tools to learner age; supervise water, cutting tools, electrical equipment, pumps, nutrient solutions and instruments; require handwashing after garden work; prohibit tasting unless a separate approved food-safety activity explicitly permits it; stop any activity when conditions are unsafe.

Before class	Check site, weather/working conditions, tools, labels, water, access routes and learner grouping.
During class	Demonstrate first; keep learners within assigned zones; monitor tool use and protect plants, structures and instruments.
After class	Count/clean/store tools; close water systems; restore labels; record damage, abnormal readings or incidents; wash hands.
Research integrity	Never invent, overwrite or silently delete observations. Flag questionable data and document repeat/reference checks.
Accessibility	Use peer roles, large-print/visual instructions, reachable observation stations and alternative recording methods where appropriate.

GARDEN 01 - Production Garden

Teaching focus: establishment, growth, yield and resource use.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Production Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-301 - QUESTION - Production Garden

Learning objective	Enable students to question a scientific investigation of establishment, growth, yield and resource use, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Production Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students question and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-302 - HYPOTHESIZE - Production Garden

Learning objective	Enable students to hypothesize a scientific investigation of establishment, growth, yield and resource use, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Production Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students hypothesize and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-303 - TEST - Production Garden

Learning objective	Enable students to test a scientific investigation of establishment, growth, yield and resource use, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Production Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students test and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-304 - REPLICATE - Production Garden

Learning objective	Enable students to replicate a scientific investigation of establishment, growth, yield and resource use, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Production Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students replicate and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-305 - MEASURE - Production Garden

Learning objective	Enable students to measure a scientific investigation of establishment, growth, yield and resource use, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Production Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students measure and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-306 - QUANTIFY - Production Garden

Learning objective	Enable students to quantify a scientific investigation of establishment, growth, yield and resource use, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Production Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students quantify and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-307 - ANALYZE - Production Garden

Learning objective	Enable students to analyze a scientific investigation of establishment, growth, yield and resource use, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Production Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students analyze and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-308 - DIAGNOSE - Production Garden

Learning objective	Enable students to diagnose a scientific investigation of establishment, growth, yield and resource use, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Production Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students diagnose and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-309 - MANAGE - Production Garden

Learning objective	Enable students to manage a scientific investigation of establishment, growth, yield and resource use, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Production Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students manage and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-310 - RECOMMEND - Production Garden

Learning objective	Enable students to recommend a scientific investigation of establishment, growth, yield and resource use, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Production Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students recommend and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 02 - Raised Systems Garden

Teaching focus: root-zone condition, infiltration and productivity.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Raised Systems Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-311 - QUESTION - Raised Systems Garden

Learning objective	Enable students to question a scientific investigation of root-zone condition, infiltration and productivity, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Raised Systems Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students question and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-312 - HYPOTHESIZE - Raised Systems Garden

Learning objective	Enable students to hypothesize a scientific investigation of root-zone condition, infiltration and productivity, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Raised Systems Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students hypothesize and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-313 - TEST - Raised Systems Garden

Learning objective	Enable students to test a scientific investigation of root-zone condition, infiltration and productivity, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Raised Systems Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students test and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-314 - REPLICATE - Raised Systems Garden

Learning objective	Enable students to replicate a scientific investigation of root-zone condition, infiltration and productivity, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Raised Systems Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students replicate and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-315 - MEASURE - Raised Systems Garden

Learning objective	Enable students to measure a scientific investigation of root-zone condition, infiltration and productivity, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Raised Systems Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students measure and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-316 - QUANTIFY - Raised Systems Garden

Learning objective	Enable students to quantify a scientific investigation of root-zone condition, infiltration and productivity, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Raised Systems Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students quantify and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-317 - ANALYZE - Raised Systems Garden

Learning objective	Enable students to analyze a scientific investigation of root-zone condition, infiltration and productivity, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Raised Systems Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students analyze and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-318 - DIAGNOSE - Raised Systems Garden

Learning objective	Enable students to diagnose a scientific investigation of root-zone condition, infiltration and productivity, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Raised Systems Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students diagnose and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-319 - MANAGE - Raised Systems Garden

Learning objective	Enable students to manage a scientific investigation of root-zone condition, infiltration and productivity, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Raised Systems Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students manage and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-320 - RECOMMEND - Raised Systems Garden

Learning objective	Enable students to recommend a scientific investigation of root-zone condition, infiltration and productivity, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Raised Systems Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students recommend and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 03 - Grid Experiment Garden

Teaching focus: fair comparisons, treatments, controls and replication.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Grid Experiment Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-321 - QUESTION - Grid Experiment Garden

Learning objective	Enable students to question a scientific investigation of fair comparisons, treatments, controls and replication, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Grid Experiment Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students question and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-322 - HYPOTHESIZE - Grid Experiment Garden

Learning objective	Enable students to hypothesize a scientific investigation of fair comparisons, treatments, controls and replication, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Grid Experiment Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students hypothesize and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-323 - TEST - Grid Experiment Garden

Learning objective	Enable students to test a scientific investigation of fair comparisons, treatments, controls and replication, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Grid Experiment Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students test and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-324 - REPLICATE - Grid Experiment Garden

Learning objective	Enable students to replicate a scientific investigation of fair comparisons, treatments, controls and replication, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Grid Experiment Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students replicate and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-325 - MEASURE - Grid Experiment Garden

Learning objective	Enable students to measure a scientific investigation of fair comparisons, treatments, controls and replication, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Grid Experiment Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students measure and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-326 - QUANTIFY - Grid Experiment Garden

Learning objective	Enable students to quantify a scientific investigation of fair comparisons, treatments, controls and replication, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Grid Experiment Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students quantify and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-327 - ANALYZE - Grid Experiment Garden

Learning objective	Enable students to analyze a scientific investigation of fair comparisons, treatments, controls and replication, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Grid Experiment Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students analyze and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-328 - DIAGNOSE - Grid Experiment Garden

Learning objective	Enable students to diagnose a scientific investigation of fair comparisons, treatments, controls and replication, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Grid Experiment Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students diagnose and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-329 - MANAGE - Grid Experiment Garden

Learning objective	Enable students to manage a scientific investigation of fair comparisons, treatments, controls and replication, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Grid Experiment Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students manage and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-330 - RECOMMEND - Grid Experiment Garden

Learning objective	Enable students to recommend a scientific investigation of fair comparisons, treatments, controls and replication, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Grid Experiment Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students recommend and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 04 - Crop Rotation Garden

Teaching focus: crop families, sequence and soil cover.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Crop Rotation Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-331 - QUESTION - Crop Rotation Garden

Learning objective	Enable students to question a scientific investigation of crop families, sequence and soil cover, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Crop Rotation Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students question and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-332 - HYPOTHESIZE - Crop Rotation Garden

Learning objective	Enable students to hypothesize a scientific investigation of crop families, sequence and soil cover, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Crop Rotation Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students hypothesize and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-333 - TEST - Crop Rotation Garden

Learning objective	Enable students to test a scientific investigation of crop families, sequence and soil cover, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Crop Rotation Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students test and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-334 - REPLICATE - Crop Rotation Garden

Learning objective	Enable students to replicate a scientific investigation of crop families, sequence and soil cover, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Crop Rotation Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students replicate and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-335 - MEASURE - Crop Rotation Garden

Learning objective	Enable students to measure a scientific investigation of crop families, sequence and soil cover, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Crop Rotation Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students measure and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-336 - QUANTIFY - Crop Rotation Garden

Learning objective	Enable students to quantify a scientific investigation of crop families, sequence and soil cover, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Crop Rotation Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students quantify and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-337 - ANALYZE - Crop Rotation Garden

Learning objective	Enable students to analyze a scientific investigation of crop families, sequence and soil cover, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Crop Rotation Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students analyze and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-338 - DIAGNOSE - Crop Rotation Garden

Learning objective	Enable students to diagnose a scientific investigation of crop families, sequence and soil cover, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Crop Rotation Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students diagnose and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-339 - MANAGE - Crop Rotation Garden

Learning objective	Enable students to manage a scientific investigation of crop families, sequence and soil cover, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Crop Rotation Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students manage and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-340 - RECOMMEND - Crop Rotation Garden

Learning objective	Enable students to recommend a scientific investigation of crop families, sequence and soil cover, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Crop Rotation Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students recommend and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 05 - Intercropping Garden

Teaching focus: component crops, canopy, yield and land-use efficiency.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Intercropping Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-341 - QUESTION - Intercropping Garden

Learning objective	Enable students to question a scientific investigation of component crops, canopy, yield and land-use efficiency, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Intercropping Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students question and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-342 - HYPOTHESIZE - Intercropping Garden

Learning objective	Enable students to hypothesize a scientific investigation of component crops, canopy, yield and land-use efficiency, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Intercropping Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students hypothesize and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-343 - TEST - Intercropping Garden

Learning objective	Enable students to test a scientific investigation of component crops, canopy, yield and land-use efficiency, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Intercropping Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students test and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-344 - REPLICATE - Intercropping Garden

Learning objective	Enable students to replicate a scientific investigation of component crops, canopy, yield and land-use efficiency, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Intercropping Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students replicate and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-345 - MEASURE - Intercropping Garden

Learning objective	Enable students to measure a scientific investigation of component crops, canopy, yield and land-use efficiency, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Intercropping Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students measure and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-346 - QUANTIFY - Intercropping Garden

Learning objective	Enable students to quantify a scientific investigation of component crops, canopy, yield and land-use efficiency, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Intercropping Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students quantify and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-347 - ANALYZE - Intercropping Garden

Learning objective	Enable students to analyze a scientific investigation of component crops, canopy, yield and land-use efficiency, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Intercropping Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students analyze and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-348 - DIAGNOSE - Intercropping Garden

Learning objective	Enable students to diagnose a scientific investigation of component crops, canopy, yield and land-use efficiency, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Intercropping Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students diagnose and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-349 - MANAGE - Intercropping Garden

Learning objective	Enable students to manage a scientific investigation of component crops, canopy, yield and land-use efficiency, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Intercropping Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students manage and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-350 - RECOMMEND - Intercropping Garden

Learning objective	Enable students to recommend a scientific investigation of component crops, canopy, yield and land-use efficiency, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Intercropping Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students recommend and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 06 - Water-Smart Garden

Teaching focus: infiltration, irrigation, runoff and water productivity.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Water-Smart Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-351 - QUESTION - Water-Smart Garden

Learning objective	Enable students to question a scientific investigation of infiltration, irrigation, runoff and water productivity, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Water-Smart Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students question and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-352 - HYPOTHEZIZE - Water-Smart Garden

Learning objective	Enable students to hypothesize a scientific investigation of infiltration, irrigation, runoff and water productivity, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Water-Smart Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students hypothesize and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-353 - TEST - Water-Smart Garden

Learning objective	Enable students to test a scientific investigation of infiltration, irrigation, runoff and water productivity, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Water-Smart Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students test and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-354 - REPLICATE - Water-Smart Garden

Learning objective	Enable students to replicate a scientific investigation of infiltration, irrigation, runoff and water productivity, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Water-Smart Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students replicate and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-355 - MEASURE - Water-Smart Garden

Learning objective	Enable students to measure a scientific investigation of infiltration, irrigation, runoff and water productivity, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Water-Smart Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students measure and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-356 - QUANTIFY - Water-Smart Garden

Learning objective	Enable students to quantify a scientific investigation of infiltration, irrigation, runoff and water productivity, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Water-Smart Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students quantify and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-357 - ANALYZE - Water-Smart Garden

Learning objective	Enable students to analyze a scientific investigation of infiltration, irrigation, runoff and water productivity, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Water-Smart Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students analyze and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-358 - DIAGNOSE - Water-Smart Garden

Learning objective	Enable students to diagnose a scientific investigation of infiltration, irrigation, runoff and water productivity, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Water-Smart Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students diagnose and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-359 - MANAGE - Water-Smart Garden

Learning objective	Enable students to manage a scientific investigation of infiltration, irrigation, runoff and water productivity, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Water-Smart Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students manage and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-360 - RECOMMEND - Water-Smart Garden

Learning objective	Enable students to recommend a scientific investigation of infiltration, irrigation, runoff and water productivity, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Water-Smart Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students recommend and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 07 - Compost & Soil Garden

Teaching focus: decomposition, maturity and soil response.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Compost & Soil Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-361 - QUESTION - Compost & Soil Garden

Learning objective	Enable students to question a scientific investigation of decomposition, maturity and soil response, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Compost & Soil Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students question and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-362 - HYPOTHESIZE - Compost & Soil Garden

Learning objective	Enable students to hypothesize a scientific investigation of decomposition, maturity and soil response, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Compost & Soil Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students hypothesize and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-363 - TEST - Compost & Soil Garden

Learning objective	Enable students to test a scientific investigation of decomposition, maturity and soil response, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Compost & Soil Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students test and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-364 - REPLICATE - Compost & Soil Garden

Learning objective	Enable students to replicate a scientific investigation of decomposition, maturity and soil response, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Compost & Soil Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students replicate and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-365 - MEASURE - Compost & Soil Garden

Learning objective	Enable students to measure a scientific investigation of decomposition, maturity and soil response, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Compost & Soil Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students measure and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-366 - QUANTIFY - Compost & Soil Garden

Learning objective	Enable students to quantify a scientific investigation of decomposition, maturity and soil response, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Compost & Soil Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students quantify and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-367 - ANALYZE - Compost & Soil Garden

Learning objective	Enable students to analyze a scientific investigation of decomposition, maturity and soil response, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Compost & Soil Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students analyze and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-368 - DIAGNOSE - Compost & Soil Garden

Learning objective	Enable students to diagnose a scientific investigation of decomposition, maturity and soil response, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Compost & Soil Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students diagnose and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-369 - MANAGE - Compost & Soil Garden

Learning objective	Enable students to manage a scientific investigation of decomposition, maturity and soil response, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Compost & Soil Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students manage and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-370 - RECOMMEND - Compost & Soil Garden

Learning objective	Enable students to recommend a scientific investigation of decomposition, maturity and soil response, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Compost & Soil Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students recommend and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 08 - Soilless Learning Garden

Teaching focus: solution flow, pH, EC and root-zone condition.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Soilless Learning Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-371 - QUESTION - Soilless Learning Garden

Learning objective	Enable students to question a scientific investigation of solution flow, pH, EC and root-zone condition, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Soilless Learning Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students question and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-372 - HYPOTHESIZE - Soilless Learning Garden

Learning objective	Enable students to hypothesize a scientific investigation of solution flow, pH, EC and root-zone condition, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Soilless Learning Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students hypothesize and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-373 - TEST - Soilless Learning Garden

Learning objective	Enable students to test a scientific investigation of solution flow, pH, EC and root-zone condition, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Soilless Learning Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students test and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-374 - REPLICATE - Soilless Learning Garden

Learning objective	Enable students to replicate a scientific investigation of solution flow, pH, EC and root-zone condition, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Soilless Learning Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students replicate and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-375 - MEASURE - Soilless Learning Garden

Learning objective	Enable students to measure a scientific investigation of solution flow, pH, EC and root-zone condition, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Soilless Learning Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students measure and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-376 - QUANTIFY - Soilless Learning Garden

Learning objective	Enable students to quantify a scientific investigation of solution flow, pH, EC and root-zone condition, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Soilless Learning Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students quantify and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-377 - ANALYZE - Soilless Learning Garden

Learning objective	Enable students to analyze a scientific investigation of solution flow, pH, EC and root-zone condition, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Soilless Learning Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students analyze and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-378 - DIAGNOSE - Soilless Learning Garden

Learning objective	Enable students to diagnose a scientific investigation of solution flow, pH, EC and root-zone condition, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Soilless Learning Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students diagnose and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-379 - MANAGE - Soilless Learning Garden

Learning objective	Enable students to manage a scientific investigation of solution flow, pH, EC and root-zone condition, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Soilless Learning Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students manage and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-380 - RECOMMEND - Soilless Learning Garden

Learning objective	Enable students to recommend a scientific investigation of solution flow, pH, EC and root-zone condition, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Soilless Learning Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students recommend and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 09 - Sensor-Ready Garden

Teaching focus: repeatability, reference checks, placement and drift.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Sensor-Ready Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-381 - QUESTION - Sensor-Ready Garden

Learning objective	Enable students to question a scientific investigation of repeatability, reference checks, placement and drift, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Sensor-Ready Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students question and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-382 - HYPOTHESIZE - Sensor-Ready Garden

Learning objective	Enable students to hypothesize a scientific investigation of repeatability, reference checks, placement and drift, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Sensor-Ready Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students hypothesize and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-383 - TEST - Sensor-Ready Garden

Learning objective	Enable students to test a scientific investigation of repeatability, reference checks, placement and drift, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Sensor-Ready Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students test and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-384 - REPLICATE - Sensor-Ready Garden

Learning objective	Enable students to replicate a scientific investigation of repeatability, reference checks, placement and drift, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Sensor-Ready Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students replicate and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-385 - MEASURE - Sensor-Ready Garden

Learning objective	Enable students to measure a scientific investigation of repeatability, reference checks, placement and drift, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Sensor-Ready Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students measure and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-386 - QUANTIFY - Sensor-Ready Garden

Learning objective	Enable students to quantify a scientific investigation of repeatability, reference checks, placement and drift, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Sensor-Ready Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students quantify and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-387 - ANALYZE - Sensor-Ready Garden

Learning objective	Enable students to analyze a scientific investigation of repeatability, reference checks, placement and drift, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Sensor-Ready Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students analyze and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-388 - DIAGNOSE - Sensor-Ready Garden

Learning objective	Enable students to diagnose a scientific investigation of repeatability, reference checks, placement and drift, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Sensor-Ready Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students diagnose and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-389 - MANAGE - Sensor-Ready Garden

Learning objective	Enable students to manage a scientific investigation of repeatability, reference checks, placement and drift, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Sensor-Ready Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students manage and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-390 - RECOMMEND - Sensor-Ready Garden

Learning objective	Enable students to recommend a scientific investigation of repeatability, reference checks, placement and drift, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Sensor-Ready Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students recommend and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

GARDEN 10 - Applied Systems Garden

Teaching focus: boundaries, stocks, flows, feedbacks and trade-offs.

Use the ten activities below as a progression rather than ten isolated exercises. Revisit earlier skills when learners need additional practice.

Garden setup	Prepare safe access, visible labels, a demonstration point and learner work zones in the Applied Systems Garden.
Journal link	Learners record observations, measurements, drawings, calculations or reflections in the level-appropriate GrowMeal journal.
Mission link	After skill practice, select a related Garden Mission Card that requires learners to apply the evidence to a challenge.
Assessment link	Use the corresponding practical, investigation/data or mission domain in the GrowMeal Assessment & Quiz System.

PA-391 - QUESTION - Applied Systems Garden

Learning objective	Enable students to question a scientific investigation of boundaries, stocks, flows, feedbacks and trade-offs, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Applied Systems Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students question and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-392 - HYPOTHESIZE - Applied Systems Garden

Learning objective	Enable students to hypothesize a scientific investigation of boundaries, stocks, flows, feedbacks and trade-offs, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Applied Systems Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students hypothesize and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-393 - TEST - Applied Systems Garden

Learning objective	Enable students to test a scientific investigation of boundaries, stocks, flows, feedbacks and trade-offs, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Applied Systems Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students test and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-394 - REPLICATE - Applied Systems Garden

Learning objective	Enable students to replicate a scientific investigation of boundaries, stocks, flows, feedbacks and trade-offs, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Applied Systems Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students replicate and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-395 - MEASURE - Applied Systems Garden

Learning objective	Enable students to measure a scientific investigation of boundaries, stocks, flows, feedbacks and trade-offs, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Applied Systems Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students measure and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-396 - QUANTIFY - Applied Systems Garden

Learning objective	Enable students to quantify a scientific investigation of boundaries, stocks, flows, feedbacks and trade-offs, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Applied Systems Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students quantify and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-397 - ANALYZE - Applied Systems Garden

Learning objective	Enable students to analyze a scientific investigation of boundaries, stocks, flows, feedbacks and trade-offs, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Applied Systems Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students analyze and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-398 - DIAGNOSE - Applied Systems Garden

Learning objective	Enable students to diagnose a scientific investigation of boundaries, stocks, flows, feedbacks and trade-offs, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Applied Systems Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students diagnose and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-399 - MANAGE - Applied Systems Garden

Learning objective	Enable students to manage a scientific investigation of boundaries, stocks, flows, feedbacks and trade-offs, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Applied Systems Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students manage and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
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Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

PA-400 - RECOMMEND - Applied Systems Garden

Learning objective	Enable students to recommend a scientific investigation of boundaries, stocks, flows, feedbacks and trade-offs, quantify response and translate the result into management reasoning.
Materials	Investigation journal, labels, measuring instruments appropriate to the garden, calculator, graph paper/spreadsheet if available, PPE where required.
DIY / low-cost option	Use calibrated marked containers, simple sampling grids, locally fabricated stakes and paper-based data sheets when digital equipment is unavailable.
Teacher preparation	Define hypothesis, variables, comparison/treatment, replicates and data-quality checks. Inspect the Applied Systems Garden and instruments before students arrive.
Step-by-step delivery	1. Frame the question and hypothesis. 2. Define independent/response/control variables. 3. Assign replicates and IDs. 4. Demonstrate instrument use and QC. 5. Students recommend and record raw data. 6. Quantify variation/effect. 7. Graph and analyze. 8. Conclude with uncertainty. 9. Recommend a management response.
Questions to ask	What is the hypothesis? Which variable is the response? Is replication adequate? What is the effect magnitude? What uncertainty qualifies the conclusion?
Expected evidence	Experimental layout, traceable raw data, processed result, graph, conclusion, limitation/uncertainty and management recommendation.
Differentiation	Provide variable templates or worked calculation examples; extension students test an alternative explanation or perform sensitivity analysis.
Troubleshooting	For suspect readings, preserve the original value, flag it, perform a reference/repeat check and document any exclusion rather than silently deleting data.
Journal / Mission / Assessment	Record the activity evidence in the student journal; follow with a related mission challenge; assess the relevant practical, data or problem-solving competency.
Close & clean up	Return tools, protect plants and equipment, restore the work area, wash hands and record any follow-up needed.

Teacher reflection: What worked? What evidence showed learning? What should change next time?

LEVEL IMPLEMENTATION REVIEW

Use this page after completing the level's practical sequence.

Strongest learner competency	
Competency needing reteaching	
Garden/system needing maintenance	
Materials to replenish	
Safety or accessibility improvement	
Next mission/assessment priority	